

Focused Hypothesis Report — Horse OLFML2A (A0A9L0SKW1) localizes to the extracellular matrix (GO:0031012)

Focus type: function_assignment (CC localization) · **Term:** GO:0031012 extracellular matrix
Target: *Equus caballus* (NCBITaxon:9796), UniProt **A0A9L0SKW1**, 599 aa, PE=4 (Predicted),
Ensembl gene model ENSECAG00000023908 / protein ENSECAP00000076704. **Human
comparison lead:** Q68BL7 (OLFML2A / Photomedin-1), 652 aa, PE=1.

Executive Judgment

**Partially supported (support via justified mammalian orthology transfer; one important
sequence caveat).**

- **Identity confirmed.** The supplied 599-aa sequence is byte-identical to the current UniProt A0A9L0SKW1 record and is an unambiguous **OLFML2A/photomedin-1 ortholog**: 87.2% identical to human OLFML2A over the mature chain, retaining the diagnostic domain architecture (N-terminal tandem CXCXCX9C cysteine motifs, mid-molecule coiled-coil, C-terminal olfactomedin domain).
- **The ortholog is genuinely an ECM protein.** Direct experimental work on the mouse ortholog (photomedin-1) shows it is **secreted, disulfide-bonded, O-glycosylated, and binds the ECM glycosaminoglycans chondroitin sulfate-E and heparin**

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P 15836428

Human OLFML2A is PE1 with UniProt CC "Secreted." This justifies ECM localization by mammalian transfer.

- **Decisive caveat on the exact sequence.** The supplied horse ORF **begins exactly at the mature-chain boundary and lacks the entire N-terminal signal peptide**. Horse residue 2 aligns to human residue 28 (the first mature residue, immediately after human SIGNAL 1..27), and the horse N-terminus has no hydrophobic h-region (max Kyte-Doolittle window-7 = 0.83 vs 3.09 in the human signal).

- **The locus DOES encode secretion-competent isoforms (isoform resolution).** Ensembl gene ENSECAG00000023908 has 6 protein-coding transcripts; the **canonical is 662 aa** (ENSECAP00000071146, N-term `MAPGTSGNLLLAACPW...`, hydrophobic, KD max 2.61) and isoform ENSECAP00000056820 (618 aa) starts `MAAAALPPRP...`, matching the human OLFML2A signal peptide (KD max 3.03). NCBI RefSeq likewise lists full-length models (XP_023484611.2, 650 aa; XP_023484612.1, 621 aa). The supplied **A0A9L0SKW1 = ENSECAP00000076704 (599 aa) is a minor, non-canonical, N-terminally-truncated isoform** — so the missing signal peptide is an **isoform/gene-model choice, not loss of secretion by the gene.**
- **Functional ECM-binding domain is intact.** The C-terminal olfactomedin (OLF) domain in the supplied 599-aa model (residues 341–599) is **99.6% identical to the human OLF domain (258/259)** and ends at the natural C-terminus (`...HFVV`). The GAG/proteoglycan-binding module that mediates ECM association is therefore fully preserved; only the N-terminal signal peptide is missing.
- **Paralog confusion ruled out.** Smith-Waterman: horse vs human OLFML2A (Q68BL7) = 983 vs human OLFML2B/photomedin-2 (Q68BL8) = 294 — unambiguously OLFML2A.
- The horse UniProt "Secreted" tag is **ARBA rule-based (automated), not experimental**, and must not be treated as biological validation.

Bottom line for the curator: ECM/secreted localization for horse OLFML2A is well supported *by orthology to an experimentally characterized ECM protein*, but is **not directly observed in horse** and is **not encoded by the supplied truncated ORF**. GO:0031012 is defensible as an ISS/ISO annotation with the signal-peptide-truncation caveat noted; it should not be asserted as directly-observed horse evidence.

Evidence Matrix

#	Citation	Evidence type	Stance	Claim tested	Key finding	Context
1	This report (computed)	Structural/ evolutionary (Smith-Waterman, own run)	Supports (identity)	Is A0A9L0SKW1 truly OLFML2A?	Horse 2–599 ↔ human 28–652, score 983, 87.2% identity ; retains OLF domain (UniProt 341–599) + N-terminal CxCxCx9C motifs + coiled-coil	Horse human protein sequence
2	This report (computed)	Computational (hydrophobicity + alignment)	Qualifies / competing	Does the exact sequence encode a signal peptide?	Horse res.2 = human res.28 (mature-chain start); no signal peptide, no hydrophobic N-term (KD max 0.83 vs human 3.09); UniProt lists no SIGNAL/TM	Exact ORF
3	PMID 15836428 (Furutani et al., 2005, <i>Biochem J</i>)	Direct assay + localization	Supports (ortholog)	Is OLFML2A/ photomedin-1 an ECM protein?	Identified in an ECM-protein screen; secreted disulfide-bonded dimer, O-glycosylated; binds chondroitin sulfate-E and heparin ; retinal photoreceptor outer segment	Mouse cDNA record protein retinal
4	UniProt Q68BL7 (human)	Review/ database	Supports (ortholog)	Human OLFML2A localization/ features	PE1; SIGNAL 1..27, CHAIN 28..652, OLF 394..652, COILED 157..183; KW Signal/ Glycoprotein/ Disulfide/Secreted; CC "Secreted"	Human
5	UniProt A0A9L0SKW1 (horse)	Database (automated)	Qualifies	Does the record itself assert ECM?	CC "Secreted" is ARBA automated ; no SIGNAL feature; no	Horse

#	Citation	Evidence type	Stance	Claim tested	Key finding	Cont
					explicit GO:0031012; PE4 Predicted	
6	Ensembl ENSECAG00000023908 + NCBI RefSeq (own queries)	Computational (isoform resolution)	Qualifies (resolves caveat)	Does the locus encode signal- peptide isoforms?	Canonical 662 aa (MAPGTSGNLLLAVCPW, KD 2.61) and isoform 618 aa (MAAAALPPRP = human signal peptide, KD 3.03) have signal peptides ; supplied 599-aa model is a minor truncated isoform (KD 0.83)	Hors chr2
7	UniProt Q68BL8 vs Q68BL7 (own SW)	Structural/ evolutionary	Supports (identity)	Is it OLFML2A not the OLFML2B paralog?	SW horse↔OLFML2A=983 vs ↔OLFML2B=294	Hors hum

Provenance: sequence identity check, Smith-Waterman alignment, N-terminal Kyte-Doolittle hydrophobicity scan, and glyc/cysteine mapping were executed in-session (code + numeric outputs retained in the iteration transcript). SHA-256 could not be recomputed in-sandbox (hashlib/struct blocked), but the provided sequence was verified byte-identical to the live UniProt A0A9L0SKW1 FASTA and length 599 matches the frozen spec.

GO Curation Implications (leads — require curator verification)

- **GO:0031012 (extracellular matrix, CC):** Biologically **defensible for the ortholog** — photomedin-1 was identified as an ECM protein and binds ECM proteoglycan GAG chains (P 15836428).

Recommend annotating by **ISO/ISS from mouse/human**, with evidence code reflecting orthology (not IDA), and a **NOT-fully-encoded-signal-peptide caveat** on the horse model.

- **Alternative/parent term consideration:** UniProt curators annotate the human protein only as "Secreted" ($\approx$ GO:0005576 extracellular region). GO:0031012 is *more specific* than the minimal database claim; it is supported by the CS-E/heparin-binding assay but a conservative curator could prefer **GO:0005576 extracellular region** if requiring the more specific ECM residence to be demonstrated for a mammalian ortholog. Either is more informative than "protein binding."
- **Supportable companion terms (leads):** MF — glycosaminoglycan binding / heparin binding (GO:0005539 / GO:0008201) from

P 15836428

CC — GO:0005576 extracellular region (parent, safest).

- **Do not** cite the horse record's ARBA "Secreted" as independent support; it is a prediction, not validation.

Mechanistic Scope

Immediate molecular/cellular function being tested is **localization/residence**, not catalysis. The gene product is a **secreted olfactomedin-domain glycoprotein** whose extracellular-matrix association is a *direct binding property* (olfactomedin/coiled-coil scaffold binding chondroitin sulfate-E- and heparin-type GAGs on proteoglycans). ECM residence is thus a primary property of the mature protein, not a downstream phenotype. The N-terminal CXCXCX9C motifs mediate disulfide-linked dimerization; the signal peptide (present in orthologs) drives secretory-pathway entry.

Conflicts and Alternatives

- **Isoform-specific truncation (principal conflict, now resolved):** the supplied ORF lacks the signal peptide, so a naïve SignalP/secretion check on the exact 599-aa sequence would be **negative**. Isoform resolution shows this is one **alternative product** of the locus: the canonical 662-aa protein and a 618-aa isoform (**MAAAALPPRP...** = human signal peptide) are secretion-competent, so the truncation is an isoform/gene-model choice, not loss of secretion by the gene.

- **Term-scope mismatch:** databases annotate the human protein as "Secreted" (extracellular region), which is broader than "extracellular matrix"; ECM specificity rests on the mouse binding assay.
- **Paralog note — resolved:** OLFML2B (photomedin-2) is a close paralog with the same architecture and ECM binding. SW scoring (horse↔OLFML2A 983 vs ↔OLFML2B 294) unambiguously assigns the horse locus to **OLFML2A**.
- **Species evidence gap:** no horse-specific localization data exist; all functional evidence is mouse/human.

Knowledge Gaps

1. **Does the true horse transcript encode a signal peptide? — largely RESOLVED.** Checked Ensembl (6 transcripts) and NCBI RefSeq: canonical 662-aa and 618-aa (MAAAAALPPRP) isoforms carry hydrophobic signal peptides; the supplied 599-aa model does not. Remaining gap: which isoform(s) predominate in horse tissue (RNA-seq/proteomics) and whether the 599-aa ORF is a real product or a mis-prediction/NMD substrate.
2. **Is ECM (vs generic extracellular) the right granularity for horse?** Checked: orthologs bind ECM GAGs (mouse). Resolve with horse tissue localization or GAG-binding assay.
3. **Locus disambiguation OLFML2A vs OLFML2B.** Resolve via synteny/phylogenetic tree of equine olfactomedin genes.

Discriminating Tests

- **Signal-peptide resolution:** SignalP-6.0 / DeepTMHMM on (a) the supplied 599-aa ORF and (b) a reconstructed full-length horse ORF from genomic + 5'-UTR; compare.
- **Transcript completeness:** map equine RNA-seq (e.g., FAANG) to ENSECAG00000023908 to confirm the true start codon and signal-peptide exon.
- **Orthology tree:** ML phylogeny of OLFML2A/OLFML2B across mammals to lock identity.
- **Localization/binding (if wet-lab):** express horse OLFML2A, test secretion and CS-E/heparin binding as in

 15836428

Curation Leads (verify before applying)

- **Reference to add:** PMID 15836428 — snippets to verify: *"characterized by the presence of tandem CXCXCX9C motifs in the N-terminal region, a coiled-coil domain and an olfactomedin domain"; "secreted as disulphide-bonded dimers (photomedin-1) ... with O-linked carbohydrate chains"; "photomedins preferentially bound to chondroitin sulphate-E and heparin."*
 - **Candidate GO action:** retain **GO:0031012** as an **ISO/ISS (orthology) annotation** with a curator note that the horse Ensembl model is signal-peptide-truncated; optionally back off to **GO:0005576** if ECM specificity for a mammalian ortholog is deemed insufficiently transferable. Consider adding **GO:0008201/GO:0005539** (heparin/GAG binding, MF) as leads.
 - **Suggested question:** Is the horse OLFML2A gene model 5'-complete? If not, flag the "Secreted"/ECM annotation as inference pending a corrected model.
 - **Do NOT** rely on the horse record's ARBA "Secreted" or agreement with predictions as validation.
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Provenance Tables (computed this run)

Signal-peptide analysis across horse OLFML2A isoforms (Ensembl ENSECAG00000023908; Kyte-Doolittle window-7):

Ensembl protein	length (aa)	status	N-terminus	KD max	KD windows >1.6	interpretation
ENSECAP00000071146	662	canonical	MAPGTSGNLLLAACPW	2.61	4	signal-peptide like (secretion-competent)
ENSECAP00000056820	618	non-canonical	MAAAALPPRPRPRP	3.03	9	= human signal peptide (secretion-competent)
ENSECAP00000036732	621	non-canonical	MDSQVFGDMDQVRM	0.83	0	NO signal peptide (truncated)
ENSECAP00000076704	599	TARGET = A0A9L0SKW1	MDSQVFGDMDQVRM	0.83	0	NO signal peptide (truncated)

Paralog disambiguation (Smith-Waterman, match +2/mismatch −1/gap −2):

Comparison	SW score	Call
horse A0A9L0SKW1 vs human OLFML2A (Q68BL7)	983	assigned
horse A0A9L0SKW1 vs human OLFML2B / photomedin-2 (Q68BL8)	294	excluded

OLF domain integrity: horse 341–599 vs human 394–652 → 258/259 = 99.6% identity; both start SCEGTLRAVDPPVRHHSYGR and end ...HFVV.

GO decision table (leads — verify):

GO id	label	aspect	recommended action	basis
GO:0031012	extracellular matrix	CC	Retain as ISO/ISS orthology annotation + caveat	Ortholog is ECM protein (P 15836428); locus has signal-peptide isoforms; supplied 599-aa model lacks signal peptide but has intact OLF domain (99.6% id)
GO:0005576	extracellular region	CC	Safer parent alternative	Matches UniProt "Secreted"; use if ECM specificity deemed insufficiently transferable
GO:0008201 / GO:0005539	heparin / GAG binding	MF	Candidate lead	Photomedin-1 binds CS-E and heparin (P 15836428)

(Provenance note: CSV export to disk was blocked by the read-only execution sandbox; tables above are the verbatim computed outputs retained from the executed code.)

Limitations: No horse-specific experimental data; functional evidence is mouse/human orthologs. Signal-peptide absence is inferred from the current Ensembl ORF and may be corrected by future gene models. SHA-256 not recomputed in-sandbox (equivalent byte-identity verified against live UniProt FASTA).